

NanoBanana Study Report: pneumothorax_cxr | stage0 | nanoba

run_dir: D:\Projects\gemini_image_segmentation\results_nanobanana\pneumothorax_cxr\gemini-3.1-fla

run_id: nanobanana_pneumothorax_smoke_desc_neg_stage0_20260302-010045

stage: stage0

dataset: pneumothorax_cxr

target: pneumothorax

model: gemini-3.1-flash-image-preview

n_tasks: 40

n_rows: 40

n_failures: 0

workers: 6

Interpretation notes:

- PRIMARY excludes duplicate/mask-source flagged retrieval runs per config.
- SENSITIVITY includes all audited rows.
- Delta tables compare each tool mode against CLOSED on paired rows.

Stall/Long-Running Events

[illegible]

Stall/Long-Running Events (cont.)

{"elapsed_seconds": 313.875, "task": "image=1.2.276.0.7230010.3.1.4.8323329.10037.1517875221.6777

Summary By Tool Mode (PRIMARY)

tool_mode	n	mean_iou	mean_dice	mean_precision	mean_recall	qc_pass_rate	iou_ci_lower	iou_ci_upper	dice_ci_lower	dice_ci_upper
closed	10.0000	0.0072	0.0135	0.0077	0.0558	1.0000	0.0000	0.0217	0.0000	0.0405
image	10.0000	0.0131	0.0247	0.0138	0.1290	0.9000	0.0000	0.0292	0.0000	0.0590
text	10.0000	0.0422	0.0661	0.0569	0.1587	1.0000	0.0025	0.1134	0.0047	0.1732
text_image	10.0000	0.0095	0.0181	0.0112	0.0695	0.9000	0.0000	0.0227	0.0000	0.0432

Summary By Tool Mode (SENSITIVITY)

tool_mode	n	mean_iou	mean_dice	mean_precision	mean_recall	qc_pass_rate	iou_ci_lower	iou_ci_upper	dice_ci_lower	dice_ci_upper
closed	10.0000	0.0072	0.0135	0.0077	0.0558	1.0000	0.0000	0.0217	0.0000	0.0405
image	10.0000	0.0131	0.0247	0.0138	0.1290	0.9000	0.0000	0.0290	0.0049	0.0589
text	10.0000	0.0422	0.0661	0.0569	0.1587	1.0000	0.0000	0.1109	0.0048	0.1637
text_image	10.0000	0.0095	0.0181	0.0112	0.0695	0.9000	0.0000	0.0227	0.0000	0.0447

Delta Vs CLOSED (PRIMARY)

tool_mode	n	mean_delta_iou	mean_delta_dice
image	10.0000	0.0058	0.0112
text	10.0000	0.0350	0.0526
text_image	10.0000	0.0023	0.0046

QC Failure Summary (PRIMARY)

tool_mode	qc_fail_count	total	qc_fail_rate
closed	0.0000	10.0000	0.0000
image	1.0000	10.0000	0.1000
text	0.0000	10.0000	0.0000
text_image	1.0000	10.0000	0.1000

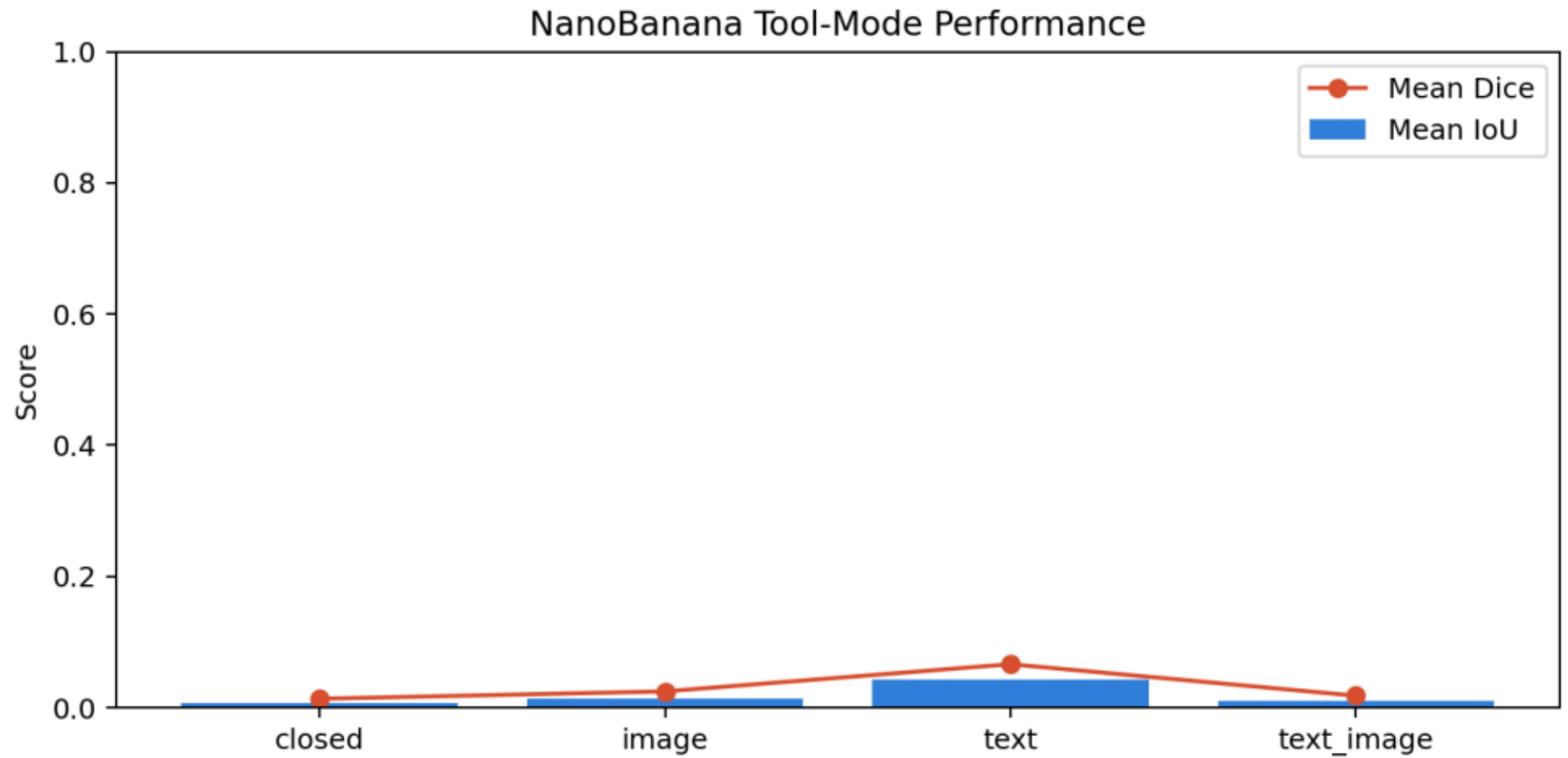
Retrieval Audit Summary

tool_mode	duplicate_rate	mask_source_rate	audit_unavailable_rate	audit_unavailable_count	n
closed	0.0000	0.0000	1.0000	10.0000	10.0000
image	0.0000	0.0000	1.0000	10.0000	10.0000
text	0.0000	0.0000	1.0000	10.0000	10.0000
text_image	0.0000	0.0000	1.0000	10.0000	10.0000

Analysis Partition Counts

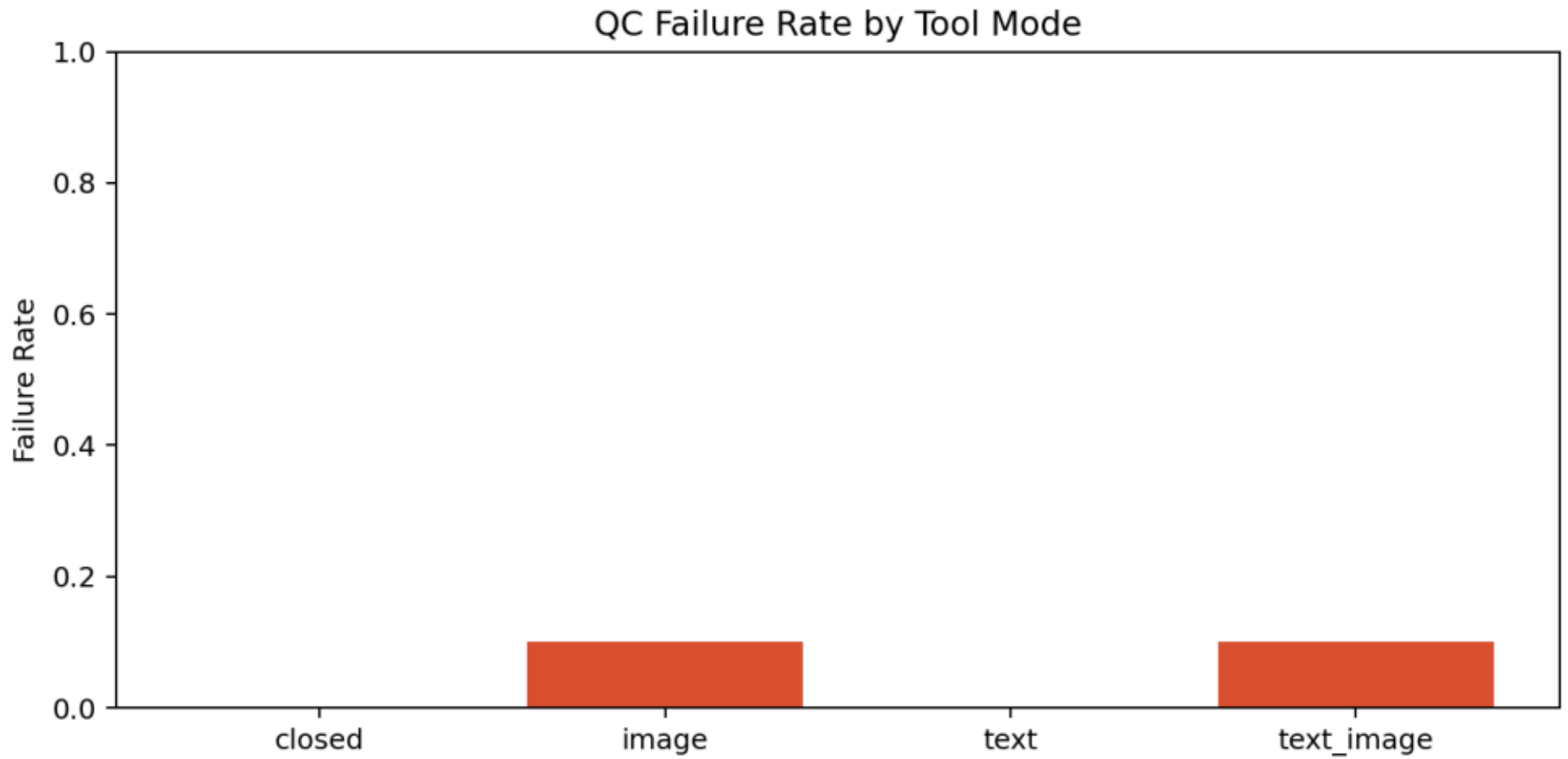
tool_mode	n_total	n_primary	n_sensitivity	n_audit_unavailable
closed	10.0000	10.0000	10.0000	10.0000
image	10.0000	10.0000	10.0000	10.0000
text	10.0000	10.0000	10.0000	10.0000
text_image	10.0000	10.0000	10.0000	10.0000

Primary Performance Plot



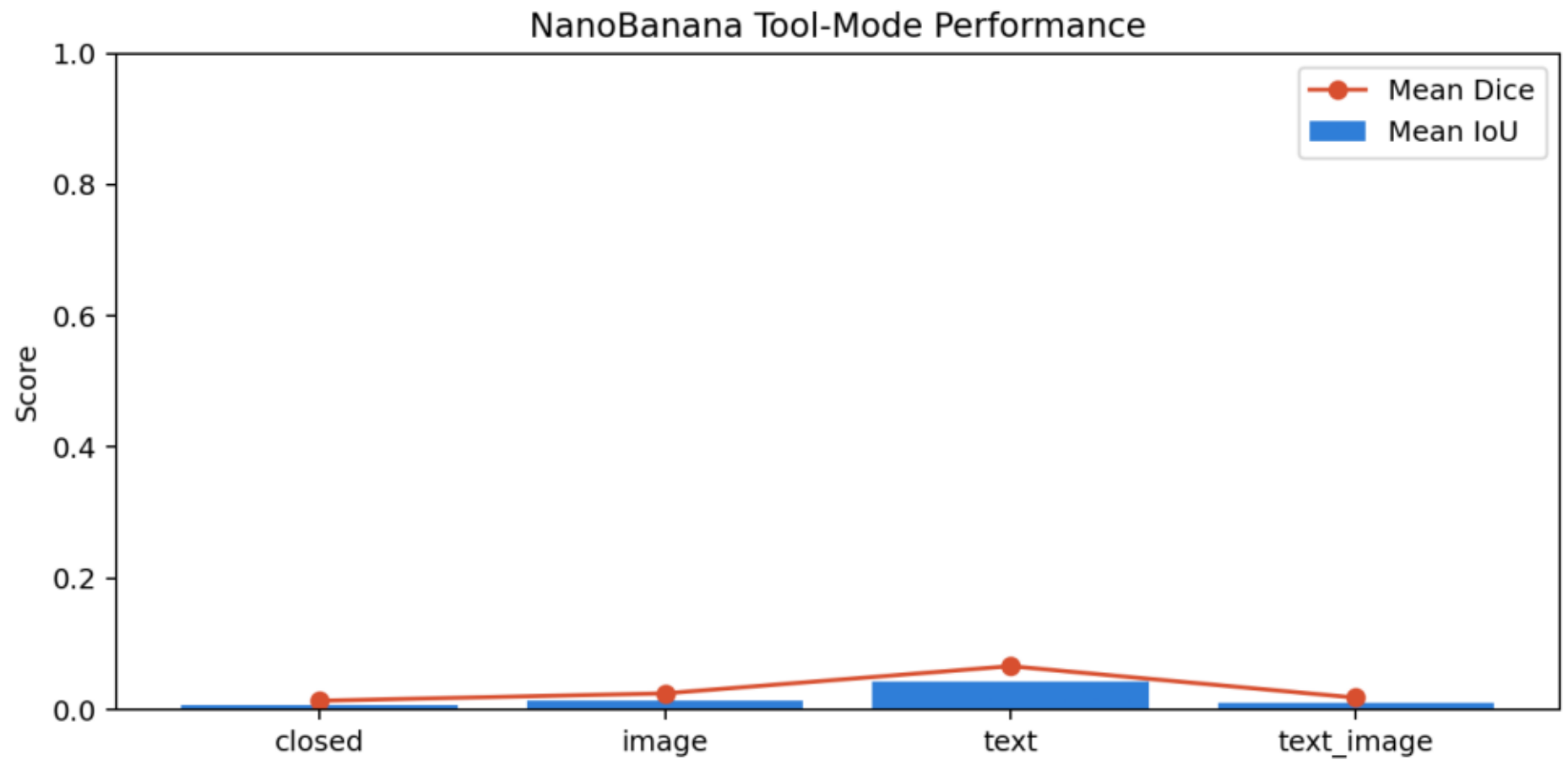
Mean IoU bars with Mean Dice line by tool mode.

Primary QC Failure Plot



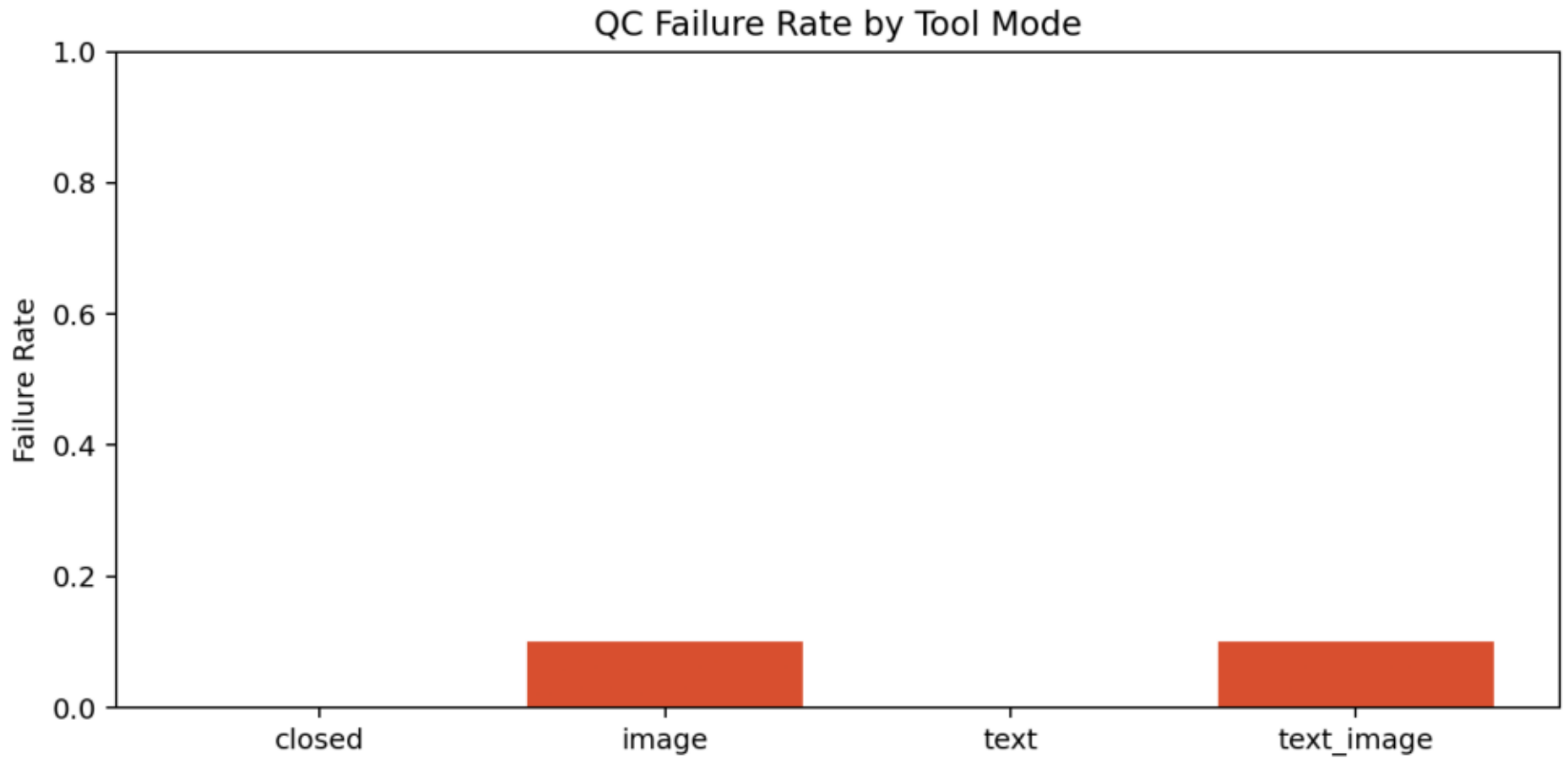
QC failure rate by tool mode.

Sensitivity Performance Plot



Sensitivity-set mean IoU/Dice by tool mode.

Sensitivity QC Failure Plot



Sensitivity-set QC failure rate by tool mode.

Per-Run Results (Excerpt)

image_name	tool_mode	iou	dice	precision	recall	qc_pass	selected_atte	retrieval_dup	retrieval_mas	analysis_primar
1.2.276.0.7230	closed	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text	0.0247	0.0482	0.0260	0.3279	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	image	0.0261	0.0509	0.0275	0.3532	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text_image	0.0000	0.0000	0.0000	0.0000	1.0000	1.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	closed	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	image	0.0000	0.0000	0.0000	0.0000	1.0000	3.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text_image	0.0000	0.0000	0.0000	0.0000	1.0000	1.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	closed	0.0723	0.1349	0.0767	0.5581	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	image	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text_image	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	closed	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	image	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text_image	0.0591	0.1116	0.0745	0.2229	1.0000	1.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	closed	0.0000	0.0000	0.0000	0.0000	1.0000	1.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text	0.3435	0.5114	0.4887	0.5362	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	image	0.0796	0.1475	0.0839	0.6102	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text_image	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	closed	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text	0.0000	0.0000	0.0000	0.0000	1.0000	1.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	image	0.0249	0.0487	0.0263	0.3265	1.0000	1.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text_image	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	closed	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	image	0.0000	0.0000	0.0000	0.0000	1.0000	1.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text_image	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000

Per-Run Results (Excerpt) (cont. 2)

image_name	tool_mode	iou	dice	precision	recall	qc_pass	selected_atte	retrieval_dup	retrieval_mas	analysis_primar
1.2.276.0.7230	closed	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	image	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text_image	0.0000	0.0000	0.0000	0.0000	1.0000	1.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	closed	0.0000	0.0000	0.0000	0.0000	1.0000	1.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text	0.0000	0.0000	0.0000	0.0000	1.0000	1.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	image	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text_image	0.0000	0.0000	0.0000	0.0000	1.0000	1.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	closed	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text	0.0536	0.1018	0.0548	0.7228	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	image	0.0000	0.0000	0.0000	0.0000	1.0000	2.0000	0.0000	0.0000	1.0000
1.2.276.0.7230	text_image	0.0361	0.0697	0.0376	0.4718	1.0000	1.0000	0.0000	0.0000	1.0000